Directed Forgetting and Second/Additional Language Learning

Morgan Chinski

Department of Linguistics, University of Illinois Urbana-Champaign

PSYC 496: Computational Foundations for Behavioral Science

Dr. Stefan Uddenberg

May 14, 2026

Directed Forgetting and Second/Additional Language Learning

Introduction

When learning a second/additional language (L2/A), we encounter an enormous amount of information in the new language. This information can align with our knowledge, or challenge our assumptions of what is correct. Opportunities where L2/A learners can notice a difference between their knowledge and the target drive error-based learning and can lead to more effective L2/A learning (VanPatten et al., 2020). However, it is still unclear in which part of the error-based learning process L2/A learners face challenges compared to native (L1) speakers: creating assumptions, or receiving information that challenges those assumptions and integrating the new information for future use (for a recent review, see Bovolenta & Marsden, 2022). Research shows that we can exert voluntary control over memory, enabling us to forget or de-prioritize specific information (see Basden & Basden, 1998; Sahakyan, 2024). This ability to intentionally forget information plays an essential role in successfully integrating feedback gained from error-based learning, thus leading to more successful L2/A learning. While previous studies have explored memory control using tasks like recognition and recall with various stimuli (Castel et al., 2011; Foster & Sahakyan, 2012; Friedman & Castel, 2011; Hennessee et al., 2017, 2018, 2019; Lee, 2013; Lo et al., 2024; Scholz et al., 2021), the role of familiarity in these processes remains largely unaddressed, especially considering familiarity with L2/A words as a factor. This study aims to explore the differences in forgetting mechanisms between English L1 and L2/A speakers when directed to forget specific words. To aim to answer my research questions, this study will use English words with different written frequencies to allow us to examine how different levels of familiarity affect cognitive processes.

Section 1 will discuss the current literature in the field of psycholinguistics, L2/A learning, word familiarity, and directed forgetting (DF). Section 2 will describe the methodology: the participants analyzed, the stimuli used, and the procedure the experiment followed. Section 3 will provide analysis of pilot data. Section 4 will draw conclusions from the data analysis, and Section 5 will provide final thoughts and conclusions.

1. Background

1.1. Directed Forgetting (DF)

Research on the voluntary control of memory have suggested that there are many factors that may play a role in how successful our attempts to intentionally forget information are. In item-method DF, words or images are followed by a Remember (R) or Forget (F) cue. Prior studies using a DF paradigm have resulted in evidence for impaired memory for items that have been cued to be forgotten by participants, known as the DF effect (Basden & Basden, 1998; Macleod, 1999; Sahakyan, 2024). It is hypothesized that this effect is due to suppression of F-items and upregulation of R-items (Bjork, 1970; Macleod, 1999). Participants attune to R-items to allow for easier recall while F-items are forgotten due to lack of rehearsal.

There has been inconsistent findings in the field of how encoding strategies are used and how the nature of stimuli affects DF. For example, the role of encoding strength has been analyzed by Geiselman et al. (1985), showing that deeply encoded items (i.e., generating synonyms for the target word) resulted in a reduced DF effect compared to shallowly encoded items (i.e., generating homonyms for the target word) due to being harder to intentionally forget. These results align with the theory that F-items exhibit DF effects due to lack of rehearsal, but other studies have found opposite effects. In Horton & Petruk (1980), subjects were tasked to either write a word that began with the same first letter, a rhyming word, or a word in the same semantic category as each designated item. The task was meant to encourage participants to create connections with the stimuli to examine if the level of encoding played a role in their ability to be intentionally forgotten. The results showed that items that were processed based on their semantic qualities (i.e., deeply encoded items) showed enhanced DF effects, instead of the predicted reduced DF compared to shallowly encoded (i.e., writing words beginning with the same letter or a rhyming word) items. The authors posited that this result was due to deeper encoding strategies disproportionately strengthening R-items and failing to benefit F-items.

Additionally, there is variation on which items are more easily forgotten than others. Lo et al. (2024) examined the effects of meaningfulness in stimuli on DF using words and nonwords. Their findings indicated a significant interaction between the meaningfulness of the stimuli and DF effect, showing that the DF effect was smaller for nonwords when compared to words. Nonwords, both R- and F-cued, also had lower average accuracy recall rates compared to meaningful words. The authors suggest that meaningfulness influences directed forgetting, and that this difference can be explained partly by the lack of episodic context in meaningless stimuli (for a review, see Sahakyan, 2024). This effect may also play a role in the performance of English L2/A learners on DF tasks. This will be further discussed in Section 1.3.

To sum up, previous experiments using DF paradigms have analyzed the effects of encoding type and nature of stimuli but have yielded inconsistent results, likely due to difference in semantic meaning in stimuli and encoding strategies. The present study aims to further explore the factors that play a role in DF, specifically word frequency, and how these factors may influence DF effects in English L2/A learners.

1.2. L2/A Learning

As discussed previously, there is disagreement where in the language learning process the deficit L2/A learners face when learning a new language occurs: failing to predict upcoming words or phrases, or failing to integrate information that has been told is outdated or incorrect (Bovolenta & Marsden, 2022). There are many aspects to language learning, but this research is concerned with prediction regarding specific words, rather than grammatical structures or violations. Many past studies have found that L2/A learners can predict upcoming words to the same extent as native speakers based on semantic context (see Dijkgraaf et al., 2017 for a review), so the present study is focused on the second theory, where learners are hypothesized to face difficulty intentionally disregarding information that they know to be incorrect. Several studies have addressed this theory to varying results (Ambridge & Brandt, 2013; Boyd & Goldberg, 2011; Henry et al., 2024; Tachihara & Goldberg, 2020, 2025). In one such study, Henry et al. (2024) found that explicit instruction combined with processing input strategies (i.e., input-oriented, see VanPatten et al., 2013) results in Spanish L2 learners scoring more accurately on an interpretation assessment using target Spanish grammatical structures. Of interest is the score on the administered delayed post-test, which showed that the group who received explicit, input-based instruction performed worse on average than every other participant group in the experiment. These findings suggest that feedback can drive error based learning in the short term, but L2/A speakers have difficulty forgetting incorrect knowledge within two weeks. Evidence of L2/A learners having difficulty forgetting explicitly corrected linguistic information has been shown in the short term as well. In a study on false cognates, Otwinowska & Szewczyk (2019) tasked Polish L1 speakers to translate English cognates, non-cognates, and false cognates into Polish and assess their confidence in the translation. Their results showed that false cognates had the highest error rate in translation, which was hypothesized by the authors to be due to participants assuming the L1 translation and being unable to disregard the association to produce the correct translation.

Prior research has also addressed the presence of false alarms on recognition tasks, which is when participants misidentify new items as being old. False alarm rates are reported in many psycholinguistic studies on the role of memory in L2/A learning, but the use of R/F cues has not been explored. In one study on error-based learning, Tachihara & Goldberg (2020) tested adult English native speakers and English L2/A learners on recognition of a mix of grammatical and ungrammatical sentences (importantly, without R/F cues). Their findings showed that the L2/A learners performed comparably to native speakers when recognizing old items, but struggled to recognize new items as being new and incorrectly indicated that new sentences had been seen before. This suggests that L2/A learners hold a bias to believing that items have been encountered before, even when they have not. In another study focusing on adult L2/A learners, English L2/A learners did not use probabilistic inferences to predict grammatical structures while native speakers did (Ambridge & Brandt, 2013), further suggesting that adult L2/A learners use different memory strategies to predict correct or acceptable sentences in the new language.

These findings demonstrate that L2/A learners can predict upcoming words, but struggle to disregard information that has been told is outdated or unnecessary to remember. Using these prior studies as a theoretical framework, it is clear that more research is needed to tease apart how this process of intentionally forgetting information differs between native speakers and L2/A learners. Word meaningfulness, as shown in Lo et al. (2024), plays a role in DF performance, but no prior research has been done examining how L2/A learners are affected by this factor. Additionally, Tachihara & Goldberg (2020) have shown that L2/A learners are more likely to misidentify new items as being old, but this topic has not been expanded to include specific cues of what information to remember or forget. For these reasons, this experiment aims to examine the role of these factors through preliminary analysis of DF performance.

1.3. Research Questions and Predictions

Previous research has not examined the interaction between word frequency and DF effects in L2/A learning. While different factors have been analyzed together, such as the role of word frequency on directed forgetting, as well as the effects of word frequency on L2/A learning, all three variables have not been looked at together. Especially considering the current discourse surrounding where in the language learning process L2/A learners encounter difficulty, this experiment aims to answer the following research questions:

1. Do English second language (ESL) learners exhibit different DF performance compared to native English speakers?
2. How does word familiarity when processing easy vs. difficult English words affect the ability to follow DF cues in both groups?

The present study predicts that English native speakers will show expected DF effects, but that English L2/A learners will show a reduced DF effect for both high and low frequency words. This prediction is supported by Lo et al. (2024), wherein Experiment 4, participants showed a reduced DF effect for meaningless stimuli. Due to their rarity in the lexicon, ESL learners may regard low frequency words as nonwords by not remembering or never learning their semantic meaning, and may show these same effects. Combined with Tachihara & Goldberg's findings (2020) that L2/A learners have a bias to incorrectly attribute new items as being old, I predict that L2/A learners in this experiment will show reduced DF effects on low and high frequency words when compared to native speakers and will have higher false alarm rates for both difficulty conditions.

2. Methods

2.1. Participants

Participants for the piloting phase of this study comprised of 14 English native speakers (*M* age = 22.89, *SD* = 1.03) and 2 English L2/A learners (*M* age = 19.00, *SD* = 0.00). Both participants in the ESL group were native Mandarin Chinese speakers currently enrolled in an ESL class at the University of Illinois Urbana-Champaign. Language proficiency for all participants was self-reported. Participants were included in the English L2/A group if they indicated on the language background questionnaire that they had taken an ESL class within the past 12 months. The study was only administered to adults over the age of 18. All participants were required to digitally consent after reading a consent form prior to their participation in the research study. After completing the online tasks, they were shown a debriefing form about the purpose of the study.

2.2. Stimuli

The stimuli included 48 words with an average Kucera-Francis frequency score of *M* = 49.9 occurrences per million words (*SD* = 15.058) ("easy") and 48 words with an average Kucera-Francis frequency score of *M* = 1.4 (*SD* = 0.47) ("difficult"). Words were chosen from the MRC Psycholinguistic Database (Coltheart, 1981) and were 5-7 letters long.

2.3. Procedure

The study was programmed using PsychoPy (Peirce et al., 2019) and hosted on Pavlovia (Bridges et al., 2020). The experiment was completed fully online. The main DF experiment had two phases: a DF phase and a recognition phase. The presentation order of the stimuli was randomized within each phase. During the DF phase, participants were presented with a mix of 24 easy and 24 difficult words one at a time. Each word was shown on the screen for 1.5 s per word, followed by an R or F cue for 3.0 s. A fixation cross was presented between each word for 1.0 s. Out of a total of 48 words, 24 were R-items and 24 were F-items (12 easy R-items, 12 easy F-items, 12 difficult R-items, 12 difficult F-items). The assignment of R/F cues was fully counterbalanced. Each word was presented in all uppercase in Arial font in the center of the screen. The R/F cue was shown slightly below where each word was presented to avoid afterimage effects. Before the DF phase, participants were instructed to remember each word presented with an R cue for a memory test later in the experiment, but that words with an F cue would not be tested and should be intentionally forgotten or disregarded.

Immediately after the DF phase, participants began the recognition phase. In this phase, participants were presented with a total of 96 words: the same 48 words from the DF phase and 48 new words. The new words were evenly split into 24 easy and 24 difficult words. Participants indicated if they had seen the presented word before or not by pressing either the “m” key on their keyboard to indicate that the word is new or the “z” key to indicate that the word was shown in the DF phase. Participants were instructed to press “z” for all words they remembered seeing previously, regardless of their R/F cue. Between each trial, a fixation cross was shown for 0.5 s.

The last task of the experiment was to complete a language background questionnaire to gain further insight into factors that may influence participants’ performance on these tasks. To qualify as an English L2/A learner, participants must have answered both that English was not their dominant language and that they had completed an ESL class within the past 12 months.

3. Descriptive Statistical Analysis

To evaluate how R-items and F-items of varying difficulty levels were recalled by both groups, the hit rate of items was calculated using the proportion of cued items successfully identified as being old for each condition (i.e., easy R-items, easy F-items, difficult R-items, and difficult F-items). Hit rate was then averaged between the participants in each group. False alarm rates, where participants incorrectly identified new items in the recognition phase as an old item, were calculated similarly by using the proportion of incorrectly recalled new items out of the total number of new items for each condition. False alarm rates were averaged between the participants in each group.

Analysis on English native speakers' hit rates of easy R-items ($M = 0.762$, $SD = 0.168$) compared to easy F-items ($M = 0.519$, $SD = 0.190$) show that English native speakers demonstrate the expected DF effects for easy words. For difficult words, English L1 participants demonstrated a higher accuracy rate on difficult R-items ($M = 0.848$, $SD = 0.187$), but had expected impaired memory performance for difficult F-items ($M = 0.728$, $SD = 0.203$).

ESL participants also demonstrated a DF effect for easy F-items ($M = 0.727$, $SD = 0.129$) compared to R-items ($M = 0.846$, $SD = 0.109$). Difficult words showed similar recall rates, with R-items having a slightly higher accuracy rate ($M = 0.709$, $SD = 0.059$) compared to F-items ($M = 0.667$, $SD = 0.112$).

False alarm rates on easy words the English native speaker group ($M = 0.063$, $SD = 0.088$) were lower than for difficult words ($M = 0.229$, $SD = 0.029$). ESL speakers showed lower false alarm rates for both easy words ($M = 0.170$, $SD = 0.180$) and difficult words ($M = 0.238$, $SD = 0.155$).

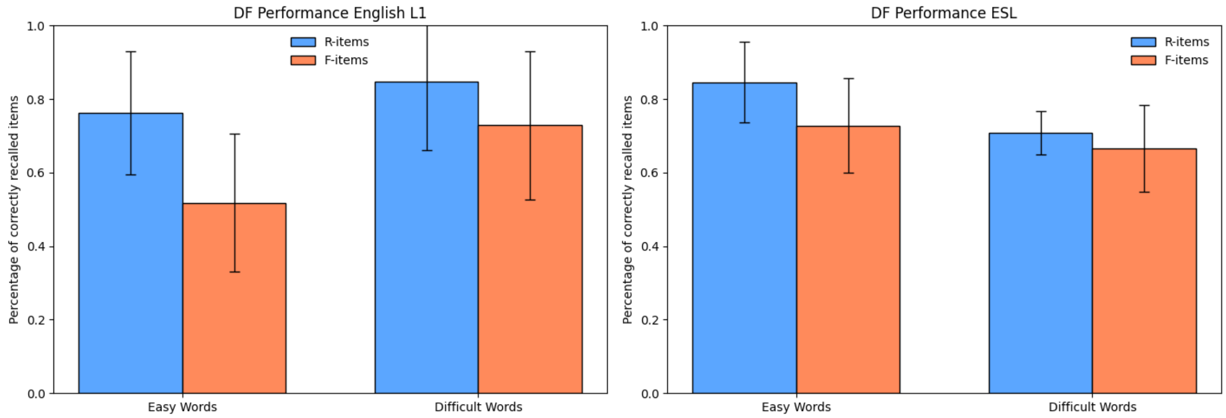

Figure 1. Comparison of the proportions of accurately recalled R- and F-items for words by level of difficulty. The left panel represents the results from the English L1 participant group. The right panel represents the results from the ESL participant group. Error bars represent the standard deviation from the mean.

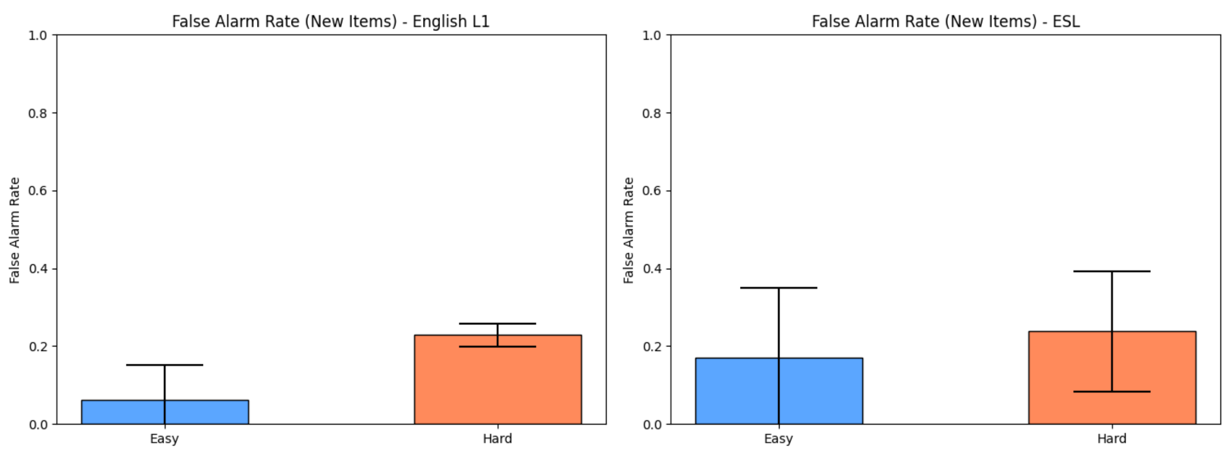

Figure 2. Comparison of the proportions of inaccurately recalled new easy and difficult items. The left panel represents the results from the English L1 participant group. The right panel represents the results from the ESL participant group. Error bars represent the standard deviation from the mean.

4. Discussion

The aim of the current study was to answer two research questions regarding the effect of word familiarity on DF performance in two groups: native English speakers and ESL learners. The first research question asked if these two groups would exhibit differences in their performance on a DF task. The second research question sought to uncover the role of word familiarity in DF performance for both English L1 and English L2/A speakers.

Regarding the first research question, the preliminary results of this experiment suggest that English native speakers and ESL learners do indeed perform differently on DF tasks. The difference in the average scores for easy R- and F-items for the English native speakers is greater than the difference of ESL participants' average scores for easy words. Difficult words showed a similar trend, where the difference between average scores for the English native speaker group was larger than the difference between scores for the ESL group. These results demonstrate that the presence of an F cue does negatively impact recall rates for both speaker groups and that ESL speakers have greater difficulty disregarding information when explicitly instructed to compared to native English speakers.

The extent of these differences are further explored by the second research question. Results from this study suggest that word difficulty does play a role in DF performance for both groups. ESL learners showed a reduced DF effect when recalling difficult words as predicted, which both aligns with this experiment's hypothesis as well as the hypothesis that L2/A learners have difficulty intentionally forgetting information. Accuracy rates in the ESL group were lower for both R and F-items in the difficult condition compared to the English L1 group, suggesting that English L2/A speakers have difficulty recalling lower-frequency words regardless of cue. An unexpected finding is that ESL participants scored higher on both easy R- and F- items compared to the native English speakers. There may be several reasons for this, one being that the participant sample size is too small to represent the larger population, but another potentially being that difficult items were not causing cognitive interference in a similar way to English L1 speakers during the DF phase, as suggested by the ESL group's lower accuracy rates for difficult words.

False alarm rates were also surprising. ESL participants incorrectly identified new items as being old less frequently than English L1 participants for both difficulty conditions, but both groups had higher false alarm rates for difficult words compared to easy words. These findings are in contrast with Tachihara & Goldberg's (2020) research stating that L2/A learners are in fact more likely to describe new items as being old than English L1 speakers. More analysis must be done to fully understand the implications of these findings. Tachihara & Goldberg's experiment did not address the role of word frequency in recall rates, so this may be an aspect that requires further experimentation as data collection is ongoing.

Some limitations of the experimental setup must be discussed to fully evaluate the results. The hypothesis posited that due to English L2/A learners potentially never learning, or being unable to recall, the semantic meaning of the difficult words in the task, participants in the ESL group would demonstrate a reduced DF effect compared to English native speakers. The current experimental design does not include an explicit measure of participants' vocabulary knowledge, so this claim cannot be directly supported by these preliminary findings. In the future, a three-option multiple choice test will be administered immediately after the DF task. 12 easy words and 12 difficult words will be randomly selected from the 96-word list used in the recognition phase of the experiment to be presented one at a time during the multiple choice test. The short exam will task participants to select the closest synonym to a shown word. By using this measure, the hypothesis stated above can be examined more directly and lead to greater understanding of the underlying reasons why these results are being produced.

5. Conclusion

Previous literature has suggested that L2/A speakers have difficulty intentionally disregarding information. To measure this ability to purposefully forget information, this experiment used a DF paradigm, along with words of different frequencies, to examine the differences between English L1 and English L2/A ability to forget items when tasked to.

The current findings support the claim that ESL speakers struggle to forget information, as evidenced by the difference in average scores between R- and F-items being lower than those for English L1 speakers. The difficulty of words affected both participant groups' DF performance, as shown by the difference in recall rates for each difficulty level. The ESL speaker group showed reduced DF effects for both easy and difficult words compared to native English speakers, which supports this experiment's hypothesis. The results of participants' false alarm, however, did not support the experiment's hypothesis. ESL speakers misidentified new items as being old more frequently than English native speakers. The results of this study require more research to be fully understood and alterations to the experimental design to gain further insight into reasons why ESL speakers exhibit reduced DF effects on low-frequency words.

References

Ambridge, B., & Brandt, S. (2013). Lisa filled water into the cup: The roles of entrenchment, pre-emption and verb semantics in German speakers' L2 acquisition of English locatives. *Zeitschrift Für Anglistik Und Amerikanistik*, *61*(3), 245–263. https://doi.org/10.1515/zaa-2013-0304

Basden, B. H., & Basden, D. R. (1998). Directed forgetting: A contrast of methods and interpretations. In *Intentional forgetting: Interdisciplinary approaches.* (pp. 139–172). Lawrence Erlbaum Associates Publishers.

Bjork, R. A. (1970). Positive forgetting: The noninterference of Items intentionally forgotten. *Journal of Verbal Learning and Verbal Behavior*, *9*(3), 255–268. https://doi.org/10.1016/S0022-5371(70)80059-7

Bovolenta, G., & Marsden, E. (2022). Prediction and error-based learning in L2 processing and acquisition: A conceptual review. *Studies in Second Language Acquisition*, *44*(5), 1384–1409. https://doi.org/10.1017/S0272263121000723

Boyd, J. K., & Goldberg, A. E. (2011). Learning what not to say: The role of statistical preemption and categorization in A-adjective production. *Language*, *87*(1), 55–83. https://doi.org/10.1353/lan.2011.0012

Bridges, D., Pitiot, A., MacAskill, M. R., & Peirce, J. W. (2020). The timing mega-study: Comparing a range of experiment generators, both lab-based and online. *PeerJ*, *8*, e9414. https://doi.org/10.7717/peerj.9414

Castel, A. D., Humphreys, K. L., Lee, S. S., Galván, A., Balota, D. A., & McCabe, D. P. (2011). The development of memory efficiency and value-directed remembering across the life span: A cross-sectional study of memory and selectivity. *Developmental Psychology*, *47*(6), 1553–1564. https://doi.org/10.1037/a0025623

Coltheart, M. (1981). The MRC Psycholinguistic Database. *The Quarterly Journal of Experimental Psychology Section A*, *33*(4), 497–505. https://doi.org/10.1080/14640748108400805

Dijkgraaf, A., Hartsuiker, R. J., & Duyck, W. (2017). Predicting upcoming information in native-language and non-native-language auditory word recognition. *Bilingualism: Language and Cognition*, *20*(5), 917–930. https://doi.org/10.1017/S1366728916000547

Foster, N. L., & Sahakyan, L. (2012). Metacognition influences item-method directed forgetting. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *38*(5), 1309–1324. https://doi.org/10.1037/a0027868

Friedman, M. C., & Castel, A. D. (2011). Are we aware of our ability to forget? Metacognitive predictions of directed forgetting. *Memory & Cognition*, *39*(8), 1448–1456. https://doi.org/10.3758/s13421-011-0115-y

Geiselman, R. E., Rabow, V. E., Wachtel, S. L., & Mackinnon, D. P. (1985). Strategy control in intentional forgetting. *Human Learning: Journal of Practical Research & Applications*, *4*(3), 169–178.

Hennessee, J. P., Castel, A. D., & Knowlton, B. J. (2017). Recognizing what matters: Value improves recognition by selectively enhancing recollection. *Journal of Memory and Language*, *94*, 195–205. https://doi.org/10.1016/j.jml.2016.12.004

Hennessee, J. P., Knowlton, B. J., & Castel, A. D. (2018). The effects of value on context-item associative memory in younger and older adults. *Psychology and Aging*, *33*(1), 46–56. https://doi.org/10.1037/pag0000202

Hennessee, J. P., Patterson, T. K., Castel, A. D., & Knowlton, B. J. (2019). Forget me not: Encoding processes in value-directed remembering. *Journal of Memory and Language*, *106*, 29–39. https://doi.org/10.1016/j.jml.2019.02.001

Henry, N., Villegas, B., & Morgan-Short, K. (2024). Explicit information and working memory in second-language acquisition of the Spanish subjunctive: A replication and extension of Fernández (2008). *Studies in Second Language Acquisition*, *46*(4), 1258–1281. https://doi.org/10.1017/S0272263124000524

Horton, K. D., & Petruk, R. (1980). Set differentiation and depth of processing in the directed forgetting paradigm. *Journal of Experimental Psychology: Human Learning and Memory*, *6*(5), 599–610. https://doi.org/10.1037/0278-7393.6.5.599

Ito, A., Corley, M., & Pickering, M. J. (2018). A cognitive load delays predictive eye movements similarly during L1 and L2 comprehension. *Bilingualism: Language and Cognition*, *21*(2), 251–264. https://doi.org/10.1017/S1366728917000050

Lee, Y.-S. (2013). Costs and benefits in item-method directed forgetting: Differential effects of encoding and retrieval. *The Journal of General Psychology*, *140*(3), 159–173. https://doi.org/10.1080/00221309.2012.750591

Lemhöfer, K., & Broersma, M. (2012). Introducing LexTALE: A quick and valid lexical test for advanced learners of English. *Behavior Research Methods*, *44*(2), 325–343. https://doi.org/10.3758/s13428-011-0146-0

Liu, D., & Zhong, S. (2016). L2 vs. L1 use of synonymy: An empirical study of synonym use/acquisition. *Applied Linguistics*, *37*(2), 239–261. https://doi.org/10.1093/applin/amu022

Lo, Y.-P., Ding, H., Whitlock, J., & Sahakyan, L. (2024). The role of prior familiarisation and meaningfulness of verbal and visual stimuli on directed forgetting. *Memory*, *32*(7), 845–862. https://doi.org/10.1080/09658211.2024.2358126

Macleod, C. M. (1999). The item and list methods of directed forgetting: Test differences and the role of demand characteristics. *Psychonomic Bulletin & Review*, *6*(1), 123–129. https://doi.org/10.3758/BF03210819

Otwinowska, A., & Szewczyk, J. M. (2019). The more similar the better? Factors in learning cognates, false cognates and non-cognate words. *International Journal of Bilingual Education and Bilingualism*, *22*(8), 974–991. https://doi.org/10.1080/13670050.2017.1325834

Peirce, J., Gray, J. R., Simpson, S., MacAskill, M., Höchenberger, R., Sogo, H., Kastman, E., & Lindeløv, J. K. (2019). PsychoPy2: Experiments in behavior made easy. *Behavior Research Methods*, *51*(1), 195–203. https://doi.org/10.3758/s13428-018-01193-y

Sahakyan, L. (2024). Current Perspectives on Directed Forgetting. In M. J. Kahana & A. D. Wagner (Eds.), *The Oxford Handbook of Human Memory, Two Volume Pack* (1st ed., pp. 1257–1277). Oxford University Press. https://doi.org/10.1093/oxfordhb/9780190917982.013.42

Scholz, S., Dutke, S., & Busch, N. A. (2021). Oscillatory correlates of intentional forgetting: The role of theta and alpha power in item-method directed forgetting. *Eneuro*, *8*(5), ENEURO.0022-21.2021. https://doi.org/10.1523/ENEURO.0022-21.2021

Stanovich, K. E., & West, R. F. (2017). *Author Recognition Test* [Dataset]. https://doi.org/10.1037/t57996-000

Tachihara, K., & Goldberg, A. E. (2020). Reduced competition effects and noisier representations in a second language. *Language Learning*, *70*(1), 219–265. https://doi.org/10.1111/lang.12375

Tachihara, K., & Goldberg, A. E. (2025). Learning unacceptability: Repeated exposure to acceptable sentences improves adult learners' recognition of unacceptable sentences. *Language Learning*, *75*(1), 77–116. https://doi.org/10.1111/lang.12660

VanPatten, B., Collopy, E., Price, J. E., Borst, S., & Qualin, A. (2013). Explicit information, grammatical sensitivity, and the first-noun principle: A cross-linguistic study in processing instruction. *The Modern Language Journal*, *97*(2), 506–527. https://doi.org/10.1111/j.1540-4781.2013.12007.x